

Linear Regression

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Linear Regression

- (Draw tipping example on board)
- Loss function $\ell_w(x, y) = (y - \langle w, x \rangle)^2$
 - Pays the square of the *residual* (draw on board)
- Resulting predictor is $y = \langle w, x \rangle$
- Use padding trick to allow line to not go through origin
 - Replace x by $[x, 1]$ and w by $[w, b]$
- Could imagine more complicated scenarios, e.g., polynomial regression (draw on board)
 - But not today

Looking closer at the loss function

- Loss function: $\sum (y_i - \langle w, x_i \rangle)^2$
- Let $A \in \mathbf{R}^{n \times d}$ and $z \in \mathbf{R}^n$ be the feature vectors and labels stacked
 - (draw on board)
- Then loss function is equivalently $\|Aw - z\|_2^2$
 - First entry of $Aw - z$ is $\langle x_1, w \rangle - y_1$, square and sum (draw on board)

The loss function is convex

- Loss fn $\|Aw - z\|_2^2 = (Aw - z)^T(Aw - z) = (w^T A^T - z^T)(Aw - z)$
 $= w^T A^T Aw - z^T Aw - w^T A^T z + z^T z$
 $= w^T A^T Aw - 2w^T A^T z + z^T z$
- Claim: if $f(x) = x^T Ax + x^T b + c$, then $\nabla f(x) = (A + A^T)x + b$
- Thus $\nabla_w \|Aw - z\|_2^2 = 2A^T Aw - 2A^T z$
- Checking the Hessian, $\nabla_w^2 \|Aw - z\|_2^2 = 2A^T A \succcurlyeq 0$
 - Why? Since $2v^T A^T Av = 2\|Av\|_2^2 \geq 0$ for any vector v
- Therefore the loss function is convex

Optimizing Least Squares

- So what if the loss function is convex?
- Setting the gradient to 0 minimizes the function
- Set $\nabla_w \|Aw - z\|_2^2 = 2A^T Aw - 2A^T z$ to be 0
- That is, find $\hat{w}$ such that $A^T A \hat{w} = A^T z$
- Could solve for $\hat{w}$ by computing $\hat{w} = (A^T A)^{-1} A^T z$
 - ...but requires $A^T A$ to be invertible
 - ...and could be slow, or imprecise if ill-conditioned
- Better to just solve the linear system $A^T A \hat{w} = A^T z$ for unknown $\hat{w}$

A note on Newton's Method

- (Not the right way to solve linear regression, but what if...)
- $w' \leftarrow w - \left(\nabla_w^2 \|Aw - z\|_2^2 \right)^{-1} \nabla_w \|Aw - z\|_2^2$
- $w' \leftarrow w - \left(2A^T A \right)^{-1} 2A^T (Aw - z)$
- $w' \leftarrow w - \left(A^T A \right)^{-1} A^T Aw + \left(A^T A \right)^{-1} A^T z$
- $w' \leftarrow w - w + \left(A^T A \right)^{-1} A^T z = \left(A^T A \right)^{-1} A^T z$
 - Assumes A is invertible
- Exactly the least-squares minimizer from the previous slide

Where did squared loss come from?

An MLE perspective

- Gaussian distribution $N(\mu, \sigma^2)$ (draw picture)
 - $f(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$
- Maximum Likelihood principle: find model parameters which maximize the probability of the observed data
- $\arg \max_{\text{model parameters}} \Pr[\text{observed data} \mid \text{model parameters}]$
- Needs some generative assumption on observed data wrt model parameters
 - i.e., $(x, y) \sim P_w$, where w are the model parameters
- A common assumption: $y = \langle w, x \rangle + z$, where $z \sim N(0, \sigma^2)$

Deriving the MLE

$$\begin{aligned}y &= \langle w, x \rangle + z, \text{ where } z \sim N(0, \sigma^2) \\ \hat{w} &= \arg \max_w \Pr[(x_1, y_1), \dots, (x_n, y_n) | w] \\ &= \arg \max_w \prod_i \Pr[(x_i, y_i) | w] \\ &= \arg \max_w \prod_i \Pr[y_i | x_i, w] \Pr[x_i | w] \\ &= \arg \max_w \prod_i \Pr[y_i | x_i, w] \\ &= \arg \max_w \log \left(\prod_i \Pr[y_i | x_i, w] \right)\end{aligned}$$

$$\begin{aligned}&= \arg \max_w \sum_i \log(\Pr[y_i | x_i, w]) \\ &\quad (\text{Note: } y_i | x_i, w \sim N(\langle w, x_i \rangle, \sigma^2)) \\ &= \arg \max_w \sum_i \log \left(\frac{1}{\sqrt{2\pi}\sigma} \exp \left(-\frac{(y_i - \langle w, x_i \rangle)^2}{2\sigma^2} \right) \right) \\ &= \arg \max_w \sum_i \log \left(\frac{1}{\sqrt{2\pi}\sigma} \right) + \log \left(\exp \left(-\frac{(y_i - \langle w, x_i \rangle)^2}{2\sigma^2} \right) \right) \\ &= \arg \max_w \sum_i -\frac{(y_i - \langle w, x_i \rangle)^2}{2\sigma^2} \\ &= \arg \min_w \sum_i (y_i - \langle w, x_i \rangle)^2\end{aligned}$$

Loss function is the squared error!

Regularization

- (Draw regression picture, with polynomial vs linear fit)
- Choosing the right model is important!
 - Sometimes simpler models are better
 - E.g., a more complex model which gets 0 training error may be worse than a simpler model which gets larger training model
- Tikhonov regularization or Ridge regression
 - $\arg \min_w \|Aw - z\|_2^2 + \lambda \|w\|_2^2$
- Lasso
 - $\arg \min_w \|Aw - z\|_2^2 + \lambda \|w\|_1$
 - Prefers *sparse* solutions